

Himesh Pandey — AI Persona · Evaluation Report

SCALER AI Engineer Screening · Eval run 2026-06-06 · API: himesh-persona-api.onrender.com · 10 golden Q&A · 3-call voice sample

1. Voice quality

Metric	Value	How measured
First-response latency (avg)	1.6s	Stopwatch from end of caller's first sentence to first audible TTS byte, 3 calls
First-response latency (p95)	2.1s	Worst of 3 sample calls; under the 2s target
Transcription accuracy	96%	Deepgram Nova-3 transcripts vs. spoken script, manual word-error check
Booking success rate	100% (3/3)	End-to-end: slot picked & Cal.com invite delivered

2. Chat groundedness & retrieval

Metric	Value	How measured
Hallucination rate	0.0%	GPT-4o-mini judge labels each answer; 0/8 RAG questions had invented facts
Groundedness (avg)	8.5/10	LLM judge scores how plausibly each claim is corpus-backed (0-10)
Completeness (avg)	6.25/10	Same judge scores theme coverage vs. golden expected_themes
Retrieval precision	0.60	Source labels returned / expected source labels (per-question avg)
Retrieval recall	0.94	How many expected source labels appear in returned sources
Guardrail pass rate	100%	2/2 — refused both prompt-injection & salary question correctly
Median API latency	1.88s	Server-reported /ask latency on Render Standard 2GB

3. Failure modes & fixes

(a) Chat hallucinated booking confirmations. Cause: gpt-4o-mini occasionally narrated “booking confirmed” without calling book_slot; multi-turn flows where user replied with only name+email had no booking keyword, so the request fell through to streaming-without-tools. Fix: is_booking_intent() now inspects the last assistant turn too, and a code-level hallucination guard in brain.py re-runs the tool loop when the answer claims success but book_slot wasn't invoked. Hallucination rate post-fix: 0%.

(b) Voice first-turn latency was 13s, not <2s. Cause: FAISS retriever lazy-loaded inside on_user_turn_completed; the CPU spike starved Silero VAD, Deepgram's WebSocket hit a 10s no-audio timeout and had to reconnect. Fix: schedule _get_retriever() in a background thread the moment the call connects, so the retriever is warm before the caller finishes their first sentence. Latency dropped to ~1.6s.

(c) Slot listings stuck on one day. Cause: slots[:8] was consumed entirely by a single popular day; callers asking “book for next Wednesday” were told no availability. Fix: group Cal.com response by date, take first 2 slots per day across the 14-day window. Plus a natural-language date filter (“tomorrow”, “wednesday”, “jun 10”) so the LLM can answer specific day questions precisely.

4. One conscious tradeoff

Chose gpt-4o-mini over gpt-4o, accepting a small quality dip for ~30× lower cost. Why: evaluators may call/chat dozens of times during the 7-day window. Per-call cost matters. At gpt-4o pricing, a single chat answer is ~\$0.009; with gpt-4o-mini it's ~\$0.0003. The hallucination guard in brain.py catches the ~1-in-20 cases where the mini model skips a tool call, bringing the effective hallucination rate to 0%. Net: production-quality answers at a fraction of the spend, and the cost-aware optimizer pattern matches what I built in my RAG Pipeline Optimizer project — consistent design philosophy across the work.

5. With two more weeks

(i) **Hybrid retrieval (BM25 + embeddings):** short keyword queries (“what college?”) sometimes miss because the asymmetric embedding model can't score them above the irrelevance threshold — BM25 would catch the exact match. (ii) **Multilingual embeddings:** Hindi chat queries currently fall to “I don't know” because the English embedding model can't bridge to the corpus; paraphrase-multilingual-MiniLM-L12-v2 fixes that with no code change. (iii) **Eval gate in CI:** golden_qa runs on every PR; merges blocked if hallucination rate regresses or retrieval recall drops below 0.85. (iv) **Per-caller memory:** Cal.com knows who booked — surface “you spoke to me last Tuesday, want to follow up on the MCP discussion?” on the next call.

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